

Problem	Possible Cause	What to Do
Nothing works when the kit is switched ON	Batteries may be weak, inserted incorrectly, or the Power JST may be loose.	Check that all batteries are inserted correctly, according to the + / - markings. Check that the battery holder is firmly connected to the Power JST. Try a fresh set of batteries.
PCB board does not receive power	Loose battery connection or incorrect battery polarity.	Switch OFF the kit. Check the battery holder and Power JST connection. Make sure the batteries are positioned correctly.
Project works sometimes and stops sometimes	A JST connector may be loose.	Push each JST connector gently but firmly into its socket. Check the wires for damage or loose pins.
IR sensor is not detecting a hand or object	The object may be too far away or the sensor sensitivity may need adjustment.	Move your hand closer to the sensor. Slowly adjust the potentiometer knob on the IR sensor until it detects the object correctly.
IR sensor keeps detecting even when nothing is nearby	Sensitivity may be set too high.	Slowly turn the potentiometer to reduce the sensor sensitivity. Test again after each small adjustment.
LDR sensor is not responding to light	The sensor may not be connected properly or the room lighting may not provide enough difference.	Check that the LDR is connected to the Sensor JST. Test it by shining a torch directly on the LDR and then covering it completely.
LED does not glow	LED connection may be loose or connected to the wrong output.	Check that the LED is connected to the connector specified in the project instructions, such as CON1 or CON2. Switch OFF the power before reconnecting it.
Buzzer does not sound	Loose connection, wrong output connector, or sensor not triggering.	Check the buzzer connection and confirm that the sensor is working. Make sure the buzzer is connected to the correct output shown in the project guide.
Motor does not rotate	Motor JST may be loose, batteries may be weak, or the motor may be connected to the wrong output.	Check the Motor JST connection to CON1 or CON2 as required. Try fresh batteries and make sure the wheels can rotate freely.
Robot moves in the wrong direction	Incorrect CON1/CON2 used.	Switch OFF the power. Move the Motor JST from CON1 to CON2, or from CON2 to CON1, as required for the project. Switch the power ON and check the direction again.
One wheel moves but the other does not	One motor connection may be loose or a wheel may be stuck.	Check both motor wires and connectors. Turn the wheels gently by hand with the power OFF and make sure nothing is blocking them.
Robot does not move even though the motors are running	Wheels may be slipping, obstructed, or incorrectly mounted.	Check that the wheels are firmly attached to the motors and that the chassis can move freely.
Water pump does not work	Pump connection may be loose, the pump may not be immersed properly, or the tube may be blocked.	Check that the pump is connected to the correct output. Make sure the pump inlet is properly immersed in water and the tube is not bent or blocked. Keep the PCB and batteries dry.
Pump runs but no water comes out	Air may be trapped in the tube or the pump may not be fully immersed.	Switch OFF the kit, place the pump properly in the water, straighten the tube, and try again.
Laser does not turn ON	Laser JST may not be connected properly.	Switch OFF the kit and check that the laser module is securely connected to the Laser JST.
Laser Security System does not trigger the buzzer	Laser reflection may not be falling correctly on the LDR.	Adjust the laser and mirror so the reflected beam falls directly on the LDR. Block the reflected beam between the mirror and LDR to test the alarm.
Sensor indicator light works, but the project does not	The sensor may be working while the output component is incorrectly connected.	Check the LED, buzzer, motor, or pump connection and make sure it is connected to the correct CON1/CON2 output for that project.
Component becomes unusually hot	Incorrect connection or short circuit may have occurred.	Switch OFF and disconnect the batteries immediately. Do not continue using the kit until all connections have been checked by an adult.
Wire or connector appears damaged	Wire may have been pulled, bent, or damaged.	Stop using that component. Do not use exposed or damaged wires. Replace the damaged part before operating the kit again.

IMPORTANT SAFETY CHECK

- Always switch OFF the power before changing connections. Never connect wires by force. Hold JST connectors by the plastic housing rather than pulling on the wires. Keep the PCB and batteries away from water.
- Never look directly into the laser beam or point the laser toward anyone's eyes. The laser module should be used only under adult supervision.

STILL NOT WORKING?

Switch OFF the kit and disconnect the batteries. Start the project again from the first step and check one connection at a time. Confirm the Power JST → Sensor JST → REV (if required) → CON1/CON2 output connection against the project diagram. If the problem remains, ask an adult to inspect the connections and components before continuing.

You Can Do It!